Forecasting Competitive Percussion Scores Based on Debut Performance

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Abstract

Indoor percussion is a judged music-and-movement activity in which ensembles perform designed shows combining percussion music with staged visual movement. Adjudicated scores are expected to rise as groups refine these shows across a season. We study whether an ensemble's debut, its first valid non-championship competitive appearance, can predict its terminal championship subtotal, the combined caption score (sum of adjudicated scoring dimensions) before timing penalties. Using a newly engineered database of 4,441 Southern California Percussion Alliance (SCPA) performances spanning nine seasons (2017–2026), we construct a cohort of 781 ensemble-seasons, where each observation represents one performing ensemble in one season. Each has both a debut and a championship result. Models are fit on 574 development observations and evaluated on 207 held-out observations from 2025–2026. Against three baselines (global median, class-level median, and debut-score identity), we evaluate Ridge regression and gradient boosting. Ridge achieves a held-out mean absolute error of 2.676 points (RMSE 3.499, $R^2 = 0.769$, Spearman $\rho = 0.892$), substantially outperforming all baselines. This result reflects predictability within the recorded adjudication system: the model forecasts a later adjudicated score from an earlier one and does not measure or explain the underlying performance quality, artistry, or season development that produced either score. Findings are further limited by championship-selection bias, observational judge assignments, and SCPA-only coverage.

Keywords

competitive percussion, marching band, performance forecasting, judged competition, longitudinal data, regression, scoring systems

1 Introduction

Competitive indoor percussion is a judged music-and-movement activity. An ensemble performs a designed show that combines percussion music with staged movement and visual composition. Judges evaluate the show across scored dimensions, called *captions*: Music and Visual address the composition and performance quality, while Music Effect and Visual Effect address how successfully the design and performance communicate and sustain an overall impact [7]. These scores are expert adjudications rather than objectively measurable outcomes: there is no distance run, time recorded, or weight lifted.

Ensembles typically refine, complete, and improve the same show as they progress from early-season regional competitions toward championships. This structure raises a natural forecasting question: how much of an ensemble's eventual championship outcome is already visible at its first competitive appearance?

The Southern California Percussion Alliance (SCPA) organizes a regional competition circuit serving Southern California. Each SCPA season runs from late winter through spring, culminating in a championship event. We use SCPA as a test case for studying the internal consistency of competitive percussion adjudication across a season.

We frame the prediction task as a test of *scoring-system predictability*: given only an ensemble's debut scores and contextual features, how well can a model reproduce the championship subtotal (the combined caption score before timing penalties) that expert adjudicators later assign? Accurate forecasting implies that debut adjudication is meaningfully continuous with championship adjudication; it does not imply that debut scores cause later performance, that the scores are artistically correct, or that the forecasts should guide consequential ensemble decisions.

The contributions of this paper are: (1) a reproducible historical SCPA scoring database built from public CompetitionSuite competition records [3, 4]; (2) a leakage-controlled debut-to-championship prediction task with chronological held-out evaluation on the 2025–2026 seasons; and (3) error analysis examining where the scoring system is most and least predictable across competitive class divisions.

2 Related Work

The relationship between judging subjectivity and competitive scoring has been studied in other adjudicated sports. Wolframm [9] examined elite dressage competitions and reported significant associations between scores and contextual variables including prior ranking, starting order, home status, and competitor nationality across seven events; the study acknowledges alternative explanations and sample-size constraints. We cite this only as evidence that structured non-performance factors have been examined in another judged sport, not as a direct analogue to indoor percussion.

Trajectory modeling of sports performance across a competitive season offers a closer methodological parallel. Dolmeta, Argiento, and Montagna [2] apply Bayesian GARCH models to intra- and inter-season shot-put trajectories, demonstrating that early-season performance can inform later-season predictions. The authors explicitly scope their framework to sports with measurable physical outcomes; SCPA scores are adjudicated constructs, not physical measurements, which limits direct methodological transfer.

Within the activity domain, Winter Guard International (WGI) operates color guard, percussion, and winds competitions [8]. Its percussion season culminates in the Percussion World Championships, providing a broader championship context for ensembles that also compete in SCPA events [6]. SCPA advertises its competitions as using "WGI format" [5]; accordingly, the current WGI percussion caption structure is directly relevant to interpreting SCPA scores. However, the available sources do not establish that SCPA used unchanged, identical score sheets in every season from 2017 through 2026.

bandScores [1] uses regional score timing to discuss championship advancement in WGI color guard. That nonacademic article concerns WGI color guard, whereas the present data contain SCPA percussion performances only; it is an informal forecasting example, not peer-reviewed evidence or external validation.

This paper introduces a newly engineered competitive-percussion dataset and applies debut-to-championship prediction within it. The contribution is the dataset and the application; the regression and boosting techniques used are standard and not claimed as novel.

3 Dataset and Exploratory Analysis

Data were collected from public CompetitionSuite recap pages (a web platform through which judges submit scores at SCPA events) using the cs-parse library [3, 4]. The resulting database contains 4,441 performance records and 56,910 score records spanning the 2017–2026 SCPA seasons, excluding 2021, when no season was held. Each performance record represents one ensemble's scored appearance at a competition. Music, Visual, Music Effect, and Visual Effect are judged separately; championship finals may use multiple adjudicators for each caption. Ensembles compete within a *class*: a competitive division grouping programs by skill level. Independent ensembles compete in A (basic concepts and skills), Open (intermediate), or World (advanced) classes (coded PIA, PIO, PIW); scholastic ensembles, which are school-affiliated programs, compete in parallel scholastic classes (PSA, PSO, PSW) or the Junior division for middle-school programs (PSJ, also evaluated on basic concepts and skills similar to A class).

Three concert classes (PSCA, PSCO, PSCW) use a stationary, non-marching format. They are excluded because their caption structure omits the marching visual framework and is incompatible with the four-caption marching sheet [7]. Sixteen all-zero records are retained in the public database for historical completeness but excluded from all analysis; thirteen are marching records. The seven marching classes contain 3,808 raw performance records and 52,288 score records; after removing the thirteen marching artifacts, the clean analysis scope contains 3,795 performance records. The 2020 season, curtailed before its championship, is included in descriptive analysis only. The 2026 season is complete through April 11, 2026.

An *ensemble-season* is one performing ensemble observed across one season. Its *debut* is its first valid, non-championship scored appearance within that season.

The primary modeling target is the terminal championship subtotal score (the sum of adjudicated caption scores at the furthest championship stage the ensemble reached), excluding timing penalties (deductions for timing, logistics, or rule compliance rather than artistic merit) and bounded to $[0, 100]$. Championship competition may include prelims, semifinals, and finals, so "terminal" refers to the latest stage reached. The modeling cohort requires each ensemble-season to have both a debut and a championship result, and excludes ensembles that appeared only at championships (typically visiting groups without a regional-season debut). An ensemble-season is also excluded if that same ensemble competes in multiple classes during the season, because its terminal class is not known at debut. A program may still contribute separate ensemble-seasons when it fields distinct ensembles concurrently (for example, a school entering both a Scholastic A and a Scholastic Open ensemble in the same season). After applying these rules and a requirement of at least two ensemble-seasons per class-season (needed for the percentile formula denominator), 781 ensemble-seasons remain: 574 development observations from 2017–2019 and 2022–2024, and 207 held-out observations from 2025–2026. Only ensemble-seasons with a championship result receive a target, which induces a selection effect the model cannot remove.

Descriptive scores rise strongly within most completed seasons: the median subtotal increases from roughly 70–74 points in the first competition week to approximately 87–90 points near championships. Terminal championship medians also differ substantially by class and year, motivating the use of class, timing, and competition-context features rather than treating a debut score as directly comparable across all observations. Figure 1 shows median terminal championship subtotals by class across seasons.

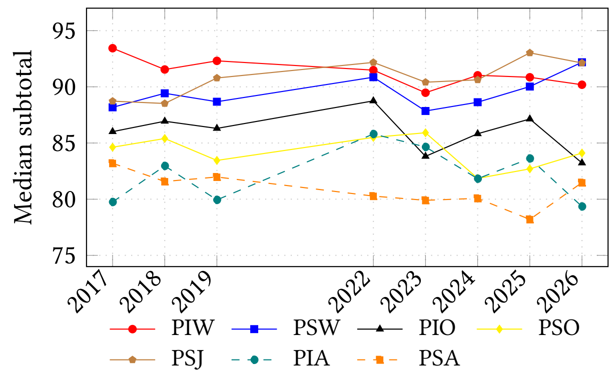

Figure 1: Median terminal championship subtotal by class, 2017–2026 (2020 and 2021 excluded because neither has a championship result). Medians vary by class and season, supporting class-aware modeling.

4 Predictive Task and Methods

Each ensemble-season in the modeling cohort represents one performing ensemble observed during one SCPA season and competing in one class.

Target. The primary target is the terminal championship subtotal score on the 0–100 scale, observed at the furthest championship stage the ensemble reached (finals, semifinals, or prelims). A secondary target (the ensemble's class-relative championship percentile) is evaluated separately and not mixed into the primary comparison. All point predictions and prediction-interval endpoints are clipped to $[0, 100]$.

Features. All features are derived from the debut performance and its context, with no information from later in the season. The debut subtotal score, timing variables (week, day of week), class membership, competition context (number of ensembles in the event and competition-week index), normalized score-profile subcaptions (finer-grained sub-scores within each caption, such as composition and performance within the Music caption, standardized among ensembles evaluated by the same judge, event, class, and subcaption), and prior-history variables (previous championship subtotal, number of prior seasons) are included. Imputation, scaling, and categorical encoding are fitted inside each training fold to prevent leakage. No post-debut score, championship attendance, championship stage, or future-season information enters as a feature.

Baselines. Three primary baselines are evaluated: (1) the global training median; (2) the debut-class training median with a global fallback for unseen classes; and (3) the debut score itself as a direct point prediction. The identity baseline reveals the raw continuity of debut scores relative to championship scores. A secondary baseline that predicts the debut-class percentile directly is evaluated on the percentile target only and is not included in the primary score table.

Table 1: Held-out evaluation on 2025–2026 seasons ($n = 207$). Spearman ρ undefined for the global median baseline (constant predictions).

Model	MAE	RMSE	R^2	Spearman ρ
Global median	5.858	7.291	−0.004	—
Class median	4.767	5.932	0.336	0.568
Debut score	11.271	12.279	−1.847	0.733
Ridge	**2.676**	**3.499**	**0.769**	**0.892**
Gradient boosting	2.934	3.805	0.727	0.868

Models. Ridge regression is trained with fold-fitted imputation, standard scaling, and one-hot class encoding; the L2 penalty is tuned by development cross-validation. The selected Ridge penalty is $\alpha = 10$. A gradient boosting model is tuned over a predeclared grid; the selected model uses 200 estimators, learning rate 0.05, maximum depth 3, and minimum leaf size 10. Stochastic procedures use random seed 42.

Validation. Development validation uses an expanding-season scheme: the model trained on 2017 predicts 2018, then successively adds seasons through 2024. The headline model is selected by development cross-validation MAE (Ridge: 2.769; GBM: 2.798), not by holdout performance. The 2025–2026 seasons are held out completely and used only for final evaluation.

Ablation. Six feature sets are evaluated sequentially to quantify each block's marginal contribution: (A) debut subtotal only; (B) add debut penalty, timing, and class; (C) add competition context; (D) add normalized score profile and missingness indicators; (E) add prior-history variables; (F) add prior class and class-change direction (upward, downward, or unchanged from the previous observed season). Each block adds only the indicated features to the previous set.

5 Results

Table 1 reports held-out evaluation metrics for all models on the 2025–2026 seasons ($n = 207$). The global-median baseline achieves an MAE of 5.858 and an R^2 near zero, consistent with constant predictions. The class-median baseline substantially improves on it (MAE 4.767, Spearman $\rho = 0.568$), confirming that class membership carries predictive information beyond a global constant. The debut-score identity baseline preserves meaningful rank ordering (Spearman $\rho = 0.733$) but severely underpredicts terminal scores, with a mean residual of −11.224 points; ensembles improve considerably between debut and championship, so the raw debut score is a poor terminal-score forecast.

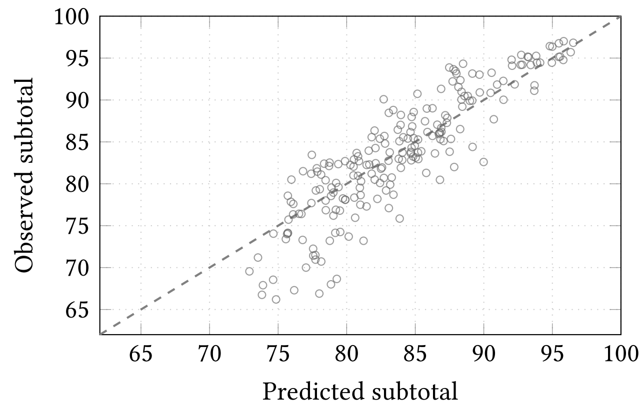

Figure 2: Ridge regression: predicted vs. observed terminal championship subtotal on the 2025–2026 held-out set ($n = 207$). The dashed line marks perfect prediction. MAE $= 2.676$, $R^2 = 0.769$.

Ridge regression achieves a held-out MAE of 2.676 (RMSE 3.499, $R^2 = 0.769$, Spearman $\rho = 0.892$). Ridge was selected as the headline model by development cross-validation MAE (2.769) rather than by post-hoc holdout comparison. A cluster bootstrap 95% confidence interval for the difference GBM MAE − Ridge MAE is $[0.056, 0.454]$ (2,000 resamples, clustered by ensemble identity), confirming that Ridge outperforms gradient boosting on this evaluation. Gradient boosting's weaker held-out performance is consistent with the structure of the task: the ablation and Ridge coefficient analysis both indicate that debut subtotal and class membership account for most of the predictive signal, leaving little residual structure for an ensemble method to exploit. At 574 development observations, the added complexity of gradient boosting does not recover additional signal over Ridge. An empirical prediction interval formed by adding and subtracting the development residual 90th percentile (5.795 points) covers 87.9% of held-out outcomes. Figure 2 shows predicted versus observed subtotals for the Ridge model.

Table 2 reports development cross-validation MAE for each ablation step. Adding debut timing and class

Table 2: Ablation results: Ridge development CV MAE as feature blocks are added sequentially. Each block adds only the indicated features to the previous set.

Feature set	Features	Dev CV MAE
A: debut subtotal only	1	3.551
B: + timing and class	6	3.060
C: + competition context	10	2.876
D: + score profile	26	2.921
E: + prior history	31	**2.798**
F: + class-change direction	33	2.798

Table 3: Class-level Ridge error; median residual separates bias from dispersion.

Class	n	MAE	Mean bias	Median resid.
PIA (Ind. A)	6	1.158	+0.408	+0.542
PIO (Ind. Open)	2	1.579	+0.384	+0.384
PIW (Ind. World)	14	1.402	+0.563	+0.176
PSA (Sch. A)	112	3.254	+0.512	+0.087
PSJ (Sch. Junior)	17	2.425	−0.990	−1.398
PSO (Sch. Open)	41	2.451	+0.348	+0.387
PSW (Sch. World)	15	1.196	−0.356	−0.580

(B) produces the largest single-step improvement. Competition context (C) adds further gain; the score-profile block (D) does not improve over C at this sample size; prior history (E) recovers the gap; and explicit class-change direction (F) adds no further improvement. Figure 3 shows the 15 largest Ridge coefficients by absolute magnitude. PIA (the lowest-tier independent class) is the regression reference category; this choice scales coefficient magnitudes relative to that baseline but does not affect model predictions or any reported metric. Debut subtotal and class membership are the dominant predictors; prior championship subtotal adds incremental signal.

Table 3 disaggregates held-out MAE and residuals by class. PSA (Scholastic A) has the highest class-level MAE (3.254 across 112 held-out observations, 54.1% of the holdout). Its near-zero median residual (+0.087), widest terminal-score (SD 6.764) and residual (SD 4.101) distributions, and large errors in both directions indicate heterogeneous development rather than systematic overprediction. PSA supplies eight of the ten largest positive residuals, partly reflecting its sample size.

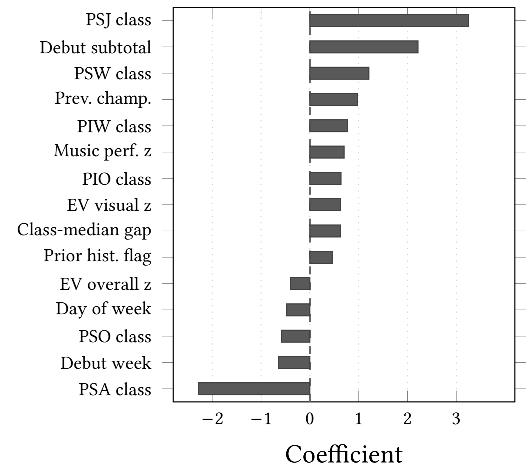

Figure 3: Top 15 Ridge regression coefficients by absolute magnitude. Class coefficients are relative to PIA, the omitted reference category; reference class choice affects coefficient scale but not model predictions. Positive values indicate higher predicted championship subtotal; negative indicate lower. Debut subtotal and class membership dominate; prior championship subtotal adds incremental signal.

6 Secondary Analysis

The secondary analysis asks whether the level of adjudicated championship scores shifted systematically across seasons, independent of the predictive task studied above. We test this by regressing championship subtotal on season indicators while controlling for class, SCPA competition week, event stage (prelims, semifinals, finals), and ensemble identity, with standard errors clustered by ensemble. Season coefficients estimate how much each season's score level differed from 2017, the omitted baseline, after these controls.

Two seasons differ significantly from 2017: 2019 (+1.113 points, 95% CI $[0.374, 1.853]$, $p = 0.003$) and 2025 (+1.123 points, 95% CI $[0.033, 2.213]$, $p = 0.043$). All other seasons (2018, 2020, 2022–2024, and 2026) are statistically indistinguishable from 2017 at the $\alpha = 0.05$ level. Figure 4 displays these adjusted effects with confidence intervals. The two elevated seasons are not adjacent, and the data do not support a claim of monotonic post-pandemic score inflation or a documented rule change.

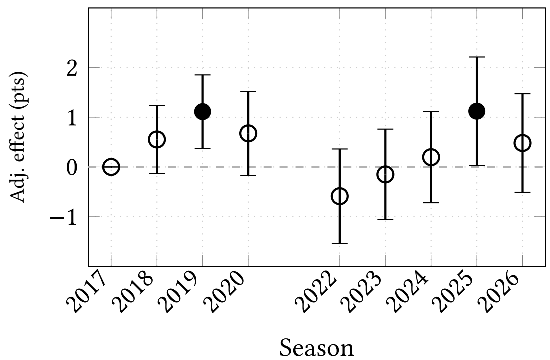

Figure 4: Adjusted season score effects relative to 2017, controlling for class, competition week, event stage, and ensemble. Error bars show 95% confidence intervals (standard errors clustered by ensemble). Filled markers indicate $p < 0.05$: 2019 (+1.11 pts) and 2025 (+1.12 pts) are statistically higher than 2017; all other seasons are indistinguishable.

7 Limitations

The principal limitation of this study is construct validity. The model observes adjudicated scores, caption-level score components, timing, class membership, competition context, and prior score history. It does not observe the performed show itself, nor most factors that shape it: musical or visual execution, artistry, instruction quality, performer membership, show design, rehearsal time, resource availability, or how fully the show had been staged and learned at debut. As a result, the model estimates continuity within the adjudication system rather than any feature of the underlying activity. A low prediction error means that debut adjudication agrees with championship adjudication; it does not mean the model explains why an ensemble improved, whether scores accurately reflect artistic quality, or how any ensemble should train or be coached. MAE measures agreement with later judges, not accuracy against an objective ground truth of performance quality.

Additional limitations include: championship-participant selection bias (only ensembles that reached a championship contribute a target); unequal schedules and sparse independent-class cells (PIO and PIA have two and six held-out observations, respectively); class changes and incomplete knowledge of intended class at debut (adjudicators may promote ensembles mid-season); debut show completeness (how fully the production has been staged at the time of the first performance), which is not recorded and may vary more across A-class ensembles than in World class, a plausible partial explanation for PSA's wider error distributions that cannot be tested with the available data; pandemic discontinuity and uncertain historical sheet revisions; and SCPA-only geographic scope, which limits generalizability to other circuits.

Judge assignments in this dataset are observational and cannot be assumed to be randomized. Because the same judges work multiple events across a regional season, scores from repeated judge–ensemble pairings are not independent. Controlling for confounders would require assignment data not available here; exploratory residual screening identified patterns that could not be attributed defensibly to individual judges under these constraints, and those findings are not presented as results.

Prediction intervals are marginal empirical bands derived from development out-of-fold residuals, not calibrated ensemble-specific guarantees.

8 Conclusion

An ensemble's debut adjudicated score meaningfully ranks later championship scores but is a poor direct forecast: the identity baseline achieves Spearman $\rho = 0.733$ but 11.271 MAE. Ridge combines debut scores, competition context, and prior history to reduce held-out error to 2.676 MAE ($R^2 = 0.769$, Spearman $\rho = 0.892$) on 2025–2026, outperforming all baselines and gradient boosting. These results measure predictability within the scoring system, not underlying performance quality. Future work should add WGI percussion results for external validation; the current database contains only SCPA performances. A prospective 2027 forecast would also provide a direct real-world test. SCPA's publicly available score records begin in 2000 [5]; incorporating pre-2017 seasons would require reconciling historical scoring systems and older recap formats.

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